GAUTENG G13 / CHALLENGE G12

SAFEGUARD SA

From identified danger to verified protection.

A shared safety-case platform for public infrastructure, restaurants, shops, workplaces and healthcare facilities.



Hackathon group concept document | 16 October 2026

Prototype proposal, not a deployed government system

01 / EXECUTIVE OVERVIEW

What is SafeGuard SA?

SafeGuard SA is a proposed mobile-friendly safety management platform that connects community hazard reports and facility inspection findings to one traceable process: risk review, temporary protection, corrective work, qualified verification and future prevention.

The problem in one sentence

A safety risk can be reported or identified in an inspection while people remain exposed, responsibility is unclear and no verified corrective outcome is visible.

What we solve

The gap between **knowing about a hazard** and proving that the hazard has been made safer.

What we create

A **shared safety-case record** linking reports, inspections, responsible people, protection, fixes and sign-off.

Who benefits

Residents, workers, customers, patients, community workers, managers and qualified inspectors.

The product promise

A report is not protection. A completed task is not automatically a safe result. SafeGuard SA makes these distinctions visible.

Official hackathon fit

The entry is anchored to **G12 - failing and distressed infrastructure**, with an initial public-hazard use case. Restaurant, retail and hospital inspections are part of the long-term product vision and must be presented as expansions, not as proof that all regulatory inspection functions fall within G12. Gauteng listed G12 as failing/distressed infrastructure and CBDs in its 2026 State of the Province Address [1].

Current project status

Concept and hackathon prototype specification. No claim is made that government has adopted SafeGuard SA, that official inspection systems are connected, or that the software can certify safety or regulatory compliance.

The dangerous gap after a finding

What goes wrong? A resident reports an uncovered manhole; a restaurant identifies unsafe food handling; or a clinic finds an obstructed emergency exit. The issue may be recorded, but people need to know who owns the next action, whether immediate protection has been applied, and whether the correction was independently checked.

A South African example

In April 2025 Johannesburg Water warned that missing manhole covers had resulted in loss of life [2]. In October 2025 eNCA reported the death of a two-year-old in an open manhole in Khutsong; residents said they had previously raised concerns [3]. These sources illustrate the danger and concerns over follow-through; they do not establish the performance of every municipal reporting system.





Illustrative AI-generated scenes: an uncovered manhole and exposed electrical wiring. Not photographs of the cited incidents.

The underlying failure

- **Fragmented entry points:** residents, staff, managers and inspectors may record risks in different systems or paper records.
- **Unclear interim safety:** a repair queue does not show whether the hazard has been temporarily contained.
- **Weak closure evidence:** a task may be marked complete without documenting an appropriate follow-up review.
- **Reactive cycles:** repeated hazards can be difficult to recognise without an accessible case and inspection history.

Precise problem statement

South African communities and facilities need a reliable way to turn reported hazards and inspection findings into **accountable, time-bound and appropriately verified safety actions**, while preserving the legal authority of existing inspectors and regulators.

03 / SCOPE

Where SafeGuard SA operates

The long-term product serves four environments. They share the **same action-and-verification engine**, but each requires its own checklist, permissions and escalation rules.

01 / Public infrastructure

Examples: open manholes, damaged walkways, exposed electricity, sewage overflows, unstable structures.
Entry: resident or field report.
G13: G12 - hackathon focus.

02 / Restaurants & food shops

Examples: food storage, cleanliness, pest control, kitchen hazards.
Entry: manager self-check or authorised EHP finding.
Regulator: municipal environmental health.

03 / Retail & workplaces

Examples: blocked fire exits, unsafe shelving, electrical risks, unsafe handling and storage.
Entry: internal workplace inspection.
Authority: authorised labour and other relevant inspectors.

04 / Hospitals & clinics

Examples: waste handling, infection-prevention facilities, equipment service records, emergency readiness.
Entry: internal quality and facilities audit.
Regulator: OHSC for prescribed standards.

Why these belong in one product

Each environment must answer the same operational questions: *What risk exists? Who noticed it? Who is responsible? What immediate protective step was taken? What corrective task is due? Who is qualified to check the outcome? When should it be inspected again?*

Boundaries that matter

- Food-premises rules include Regulation R638 of 2018; certificates of acceptability are issued through authorised environmental health processes [5, 6].
- Workplace inspections and legal enforcement fall within the powers of authorised inspectors under South African OHS legislation [7].
- The OHSC inspects and certifies healthcare establishments; its authority is not transferred to a third-party app [4].

Design rule

Every record must state its source: **public allegation**, **voluntary internal inspection** or **authorised official finding**. These are not interchangeable.

One safety case, six actions

A safety case is the single record that connects the original report or inspection finding with everyone and every action involved in reducing the risk.

01	02	03	04	05	06
REPORT	**ASSESS**	**PROTECT**	**CORRECT**	**VERIFY**	**PREVENT**
Finding logged	Priority reviewed	Interim measure	Work recorded	Qualified check	Reinspect

Core information in every case

Field	Purpose
Source and location	Public report / internal self-check / official finding, plus place or asset.
Risk and evidence	Hazard category, time observed, photos/notes collected safely.
Responsible owner	Person/team responsible for assessment and action; due dates.
Protection record	Whether protective action has been proposed, performed and verified.
Correction and review	Work done, evidence, reviewer role and decision.
History and recurrence	Audit timestamps, reopening, repeat hazard and next inspection.

Status is not the same as safety

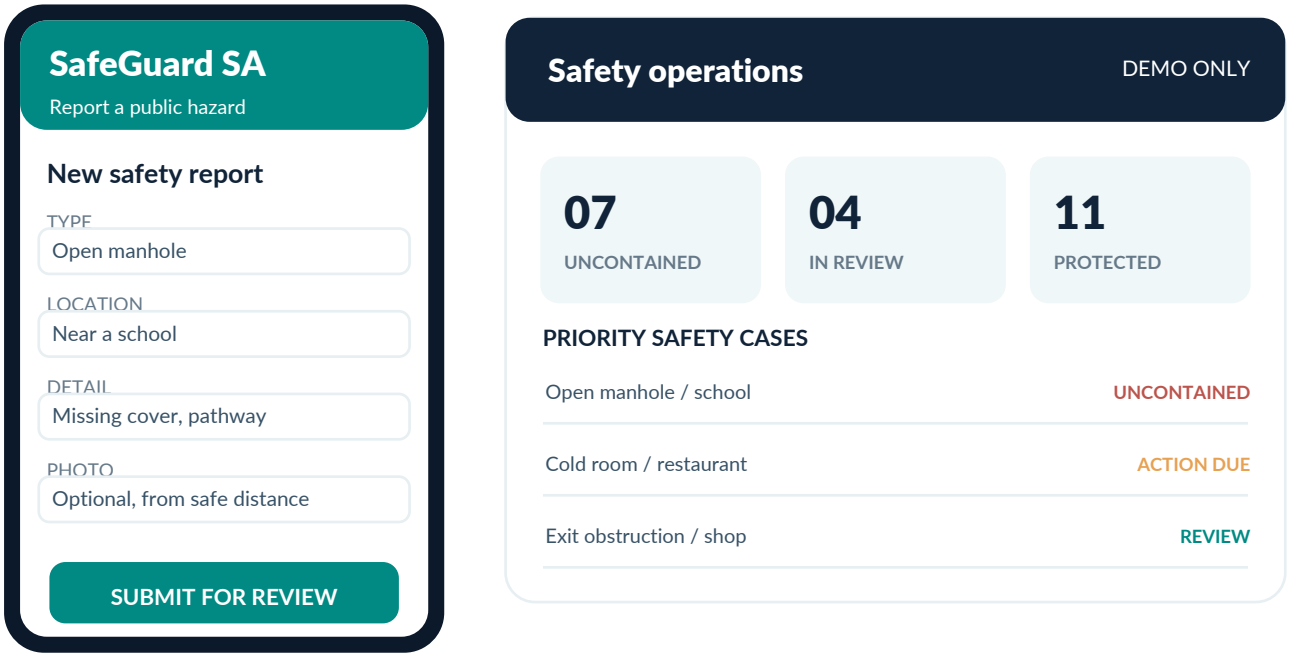
Unassessed → Under review → Uncontained → Protection recorded → Protected (verified) → Repair pending → Repair recorded → Resolved (verified). A rejected or disputed verification moves the case back into review. This is a proposed workflow, not a replacement for formal statutory processes.

Critical distinction

Recording that a barrier was installed does not prove it is adequate. The status becomes **verified protected** only when the designated qualified reviewer accepts the evidence and inspection result.

What the team is creating

SafeGuard SA would offer a **low-data resident interface** and a **role-controlled operations workspace** for inspections, corrective actions and reviews.



Above: original conceptual interface drawings; illustrative values and case details only.

Resident / public view

Simple risk category, approximate location, photo from a safe position, reference number and verified public status. No claim of automatic emergency dispatch.

Operations / inspector view

Risk queue, assigned actions, overdue cases, uploaded evidence, authorisation checks, audit history and scheduled reinspection.

Two distinct demonstration journeys

- **Public infrastructure:** open manhole → report → municipal-style triage → barrier action → verified interim protection → cover replacement → verified closure.
- **Restaurant:** cold-storage concern in a manager self-check → corrective task → staff evidence → manager recheck. An official EHP determination remains separate.

Who uses SafeGuard SA?

People have different visibility, responsibilities and legal powers. Access must be role-based from the first version.

User	Action in the platform	Limits
Resident / customer	Submit a safe-distance report; see restricted public case status.	Cannot certify, approve repairs or view private inspection files.
Community safety worker	Record observations, assist with reports, check visible hazards.	No authority to certify technical safety unless separately qualified.
Business / facility staff	Run self-checklists; receive and complete internal tasks.	Voluntary record, not a regulator inspection.
Operations manager	Review issues, assign work, set deadlines, monitor repeat findings.	Cannot make official legal compliance determinations without authority.
Field / maintenance team	Record interim actions, repairs and completion evidence.	Work recorded does not mean accepted.
Qualified reviewer / inspector	Assess evidence and site conditions within their delegated authority.	Only approved authorities issue official notices or certificates.

Responsible authorities in South Africa

- **Restaurants and food shops:** authorised municipal Environmental Health Practitioners; oversight and instruments under food-premises rules [5, 6].
- **Workplaces:** Department of Employment and Labour OHS inspectors and other competent authorities, depending on the hazard [7].
- **Hospitals and clinics:** OHSC compliance inspections and certification under applicable healthcare norms [4].
- **Public infrastructure:** the relevant municipal or asset-owning authority and competent technical inspectors.

Trust principle

The app should never generate a government certificate, claim a voluntary self-check is an official inspection, or automatically clear an electrical, structural or clinical safety issue.

07 / COMPETITIVE POSITION

What exists - and what is different?

We cannot truthfully claim to have invented safety inspections, digital checklists, fault reporting or corrective-action tracking. These capabilities already exist. The proposed advantage is an **integrated South African public-to-institutional safety pathway**, with verified interim protection and controlled regulatory boundaries.

Existing alternative	What it already provides	Proposed SafeGuard SA emphasis
City of Johannesburg reporting	Fault logging, reference numbers and service follow-up [8].	Explicit distinction between uncontained risk, verified temporary protection and verified final repair.
Fixlocal	Report and track local issues; guidance on escalation [9].	A shared operational case involving field protection, inspection and sign-off.
SafetyCulture	Checklists, audits, corrective tasks, investigations, asset and action records [10].	A public/community entry point combined with facility operations and sector-specific permissions.
EHP / labour / OHSC processes	Legally authorised inspections, notices and certification [4–7].	Supports authorised or internal follow-through; does not replace legal authority.

Four aspects we would attempt to differentiate

- **Two separate safety outcomes:** interim protection verified, and permanent correction verified.
- **One traceable record:** a public report or inspection finding can progress through responsible action and audit history.
- **Public-safe transparency:** residents see hazard status without seeing confidential business, medical or personnel records.
- **South African permissions:** a voluntary check is clearly separated from decisions reserved for EHPs, labour inspectors and the OHSC.

The honest innovation claim

SafeGuard SA is a **proposed combination and workflow emphasis**, not a verified world-first invention. Before claiming this exact function is unavailable, interview inspectors, municipalities and facility managers and test leading products against the same use case.

Why SafeGuard SA matters

The app is designed to improve **follow-through, accountability and prevention**. It cannot itself install covers, restore wiring, make buildings safe or guarantee compliance; those require real services, competent people and budgets.

For residents

Better visibility of known hazards, public warnings only when verified, and clearer reference and follow-up information.

For facilities

Repeatable inspection checklists, task ownership and evidence of corrective action between formal inspections.

For government

A clearer view of safety cases still uncontained, repeated failures, overdue work and documented close-out.

Measurable pilot indicators

Metric	What it measures
Time to initial assessment	Elapsed time from logged finding to qualified or authorised triage.
Time hazard remains uncontained	Elapsed time from verified finding to accepted interim protection.
Overdue actions	Share of assigned corrective actions beyond target date.
Verified closure rate	Proportion of recorded corrections supported by required review.
Repeat hazards	Same type of hazard recurring at a location or facility.
Checklist completion quality	Missing required evidence, signatures and required follow-ups.

Long-term relevance

Even when one hazard has been fixed, routine inspections, asset servicing, changing workplace conditions and new infrastructure risks create continuing need. The platform should move from **reactive reporting** toward **scheduled prevention**, not depend on an endless backlog of emergencies.

Evidence standard

These are intended benefits and pilot measures. Do not claim SafeGuard SA has already reduced injuries, inspection failures, government costs or response times.

09 / HACKATHON BUILD

An achievable first release

Our goal at the expo is **one credible end-to-end demonstration**, not a fully operational regulator or municipal system.

Must build (MVP)	Demonstration acceptance criterion
Create safety finding	Resident logs fictional open-manhole case; manager logs a restaurant self-check.
Role-based sign-in	Public user cannot access private operational and inspection data.
Triage + ownership	Coordinator assigns a responsible person, priority and due date.
Interim protection	Field team records proposed barrier; safety status stays unverified until authorised review.
Corrective action	Repair or facility task is logged with completion evidence.
Verification / audit	Reviewer accepts or rejects outcome; full status history is visible.
Dashboard	Shows uncontained, action due, pending verification and resolved counts.

Suggested technology

Frontend: React + Tailwind (responsive web app). **Backend:** Supabase/PostgreSQL with row-level access rules. **Hosting:** Vercel. **Files:** restricted evidence uploads. **Testing:** fictional residents, cases, facilities and inspector accounts. **Integrations:** no live government APIs required for the demo.

Five-minute demo sequence

1. Show unsafe open manhole → 2. Submit report → 3. Assign safety case → 4. Record barrier → 5. Inspector verifies protection → 6. Crew repairs → 7. Inspector closes → 8. Show restaurant self-check using the same case engine.

Cut list if time runs short

Drop maps, AI hazard classification, SMS, WhatsApp, machine-learning prediction and regulator integrations. Keep the complete verified safety-case workflow working.

10 / ROLLOUT AND GOVERNANCE

How it can become a real service

A responsible rollout starts with a narrowly supervised pilot and expands only when inspection authority, data handling and organisational responsibilities are agreed.

Stage	Scope and validation
Hackathon proof of concept	One infrastructure hazard + restaurant self-check; synthetic data, simulated roles.
Community pilot	Partner with a municipality or community safety organisation; validate risk prioritisation and case closure.
Facility self-checks	Onboard consenting businesses with fit-for-purpose checklists and qualified internal reviewers.
Healthcare extension	Co-design with facility quality teams; no patient records or regulatory claims without formal arrangements.
Regulator integration	Only through formal approval, legal assessment, interoperability standards and contracts.

Safety, privacy and responsibility

- **Emergency response:** urgent danger must display existing emergency-service guidance; reporting is not equivalent to emergency dispatch.
- **Data minimisation:** avoid children's names, patient records and unnecessary identifying images; apply privacy notices and access restrictions under POPIA [11].
- **Evidence security:** private file storage, role-based access, access logs, deletion and retention rules, and clear separation of public/internal data.
- **Human authority:** high-risk structural, electrical and clinical matters require suitably qualified staff; no automated compliance certificate.

Risks that remain outside the app

Municipal adoption, staffing, budgets, transport, actual repairs, inspection mandates, network access, false reports, duplicate submissions and the reliability of evidence. The team must test these constraints with real potential users before public rollout.

The group execution blueprint

For ten contributors, split responsibility into five pairs. Confirm the hackathon's team-size rules with an organiser before final registration or submission.

Pair	Responsibility	Immediate deliverable
1–2	Problem validation, legal context and competitor review.	Source pack, two interviews, gap evidence.
3–4	UX, accessibility, and user journeys.	Resident and safety operations flows.
5–6	Frontend and responsive screens.	Report, inspection and dashboard screens.
7–8	Database, permissions and workflow integration.	End-to-end case updates and access rules.
9–10	Quality assurance, presentation and demo.	Test report, pitch and backup recording.

Four decisions to make with mentors

- **Challenge fit:** will the judges accept a G12 entry with future food, retail and healthcare inspection modules?
- **Operational gap:** do relevant municipal departments already track protective action separately from permanent repair?
- **Review authority:** who could legitimately sign off protection and closure in a supervised pilot?
- **First adopter:** which municipality, safety-focused NGO, facility or business would be willing to test a workflow?

Recommended pitch line

"SafeGuard SA helps turn safety findings into verified protection. We are creating a shared safety-case workflow that connects community reports and facility inspections to action, responsible ownership, qualified review and recurring prevention."

What success at the expo looks like

A working prototype where a hazard is not automatically "safe" when reported, assigned or repaired, and a second facility self-check uses the same operational engine with correctly separated permissions.

Evidence and verification sources

[1] **Gauteng Provincial Government (2026)** 2026 State of the Province Address - G12 failing/distressed infrastructure and CBDs
https://www.gov.za/news/speeches/premier-panyaza-lesufi-gauteng-state-province-address-2026-24-feb-2026

[2] **Johannesburg Water (2025)** Theft and vandalism of water infrastructure; manhole public-safety warning
https://www.johannesburgwater.co.za/blog/2025/04/08/theft-and-vandalism-of-water-infrastructure-in-johannesburg-is-crippling-effective-service-delivery-2/

[3] **eNCA (2025)** Khutsong open-manhole fatality and reported prior concerns
https://www.enca.com/top-stories/outrage-over-toddlers-death-open-manhole-khutsong

[4] **Office of Health Standards Compliance** Regulator role, health-establishment inspections, certification and enforcement
https://ohsc.org.za/operations/

[5] **National Department of Health** Environmental Health: EHP inspections, follow-up and certificates
https://www.health.gov.za/eh-home/

[6] **National Department of Health** Food-premises hygiene regulations, R638 of 22 June 2018
https://www.health.gov.za/food-control-regulations/

[7] **Department of Employment and Labour** OHS inspector duties and enforcement responsibilities
https://www.labour.gov.za/ohs-inspector-%2826%29

[8] **City of Johannesburg** Report a city problem: existing municipal reporting and tracking
https://eservices.joburg.org.za/pages/Reportacityproblem.aspx

[9] **Fixlocal** Existing local issue reporting, tracking and escalation guidance
https://www.fixlocal.org.za/

[10] **SafetyCulture** Inspections, audits, tasks and corrective actions already offered
https://safetyculture.com/platform/

[11] **South African Government** Protection of Personal Information Act 4 of 2013
https://www.gov.za/documents/protection-personal-information-act

[12] **OHSC inspection report 2023/24** Illustrates formal healthcare inspection and reinspection processes
https://ohsc.org.za/wp-content/uploads/2025/11/OHSC-Approved-Annual-Inspection-Report-2023_24.pdf

Definitions to remember

Hazard: a source of potential harm. **Safety case:** the record linking a finding to risk assessment and action. **Interim protection:** a temporary step reviewed by competent staff to reduce exposure. **Corrective action:** work addressing the finding. **Verified closure:** a documented acceptance within a reviewer's legitimate authority. **Official inspection:** a legally authorised process, not merely a self-check.

Prepared as a concept-definition and planning document for the Youth Tech Expo team. All app screens, cases and planned features are illustrative; actual availability, compliance and public impact require implementation and validation.